

Simulating Logistics Systems

Lesson 1/15

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Outline of this lecture

Systems and models

Methodology

Simulation project management

Advantages and disadvantages

Software: SimPy (Python)

Python Practice

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- ▶ **Prescriptive** models (*what should happen in the future*): They are generated when information that does not exist in the current system is required (*optimization, heuristics, metaheuristics, simulation-based optimization*).

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- ▶ **Discrete Event Simulation:** DES models a system as a sequence of events that occur at distinct times. It is used to analyze processes where changes happen at specific moments, such as in queues (e.g. *patients in hospital*) [Tocher, 1963];

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*Simulation **does not replace intelligent thinking** in the decision-making process. The tool serves as a support to the decision-making process.*

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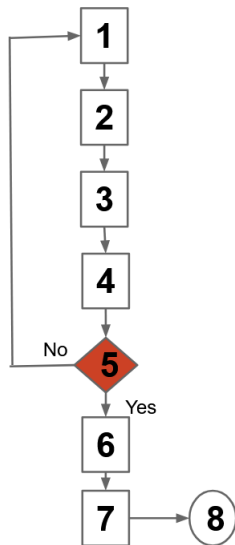
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- ▶ **Stochastic:** Contains one or more random variables.

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Failures in simulation studies [Law et al., 2007, Banks, 2005]

“Asking the right question is half the answer.” Popular saying.

1. Unclear Objective: What is really important?
2. Failure to define objectives at the beginning of the simulation study.
3. Lack of communication with the client.
4. Failure to communicate with management during the study.
5. Failure to collect good data or data with an inadequate level of detail.
6. Failure to consider all sources of randomness.
7. Analyzing results from a single replication and treating the result as the “answer.”
8. Using the wrong measures to evaluate the system.
9. Treating simulation as programming, lacking team expertise in simulation.
10. Using inadequate or poorly documented simulation software.

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Advantages [Banks, 2005, Pidd, 1998, Law et al., 2007]

- ▶ Many systems cannot be accurately evaluated analytically. Simulation is one possible method;
- ▶ It allows the performance of a system to be estimated under projected operating conditions;
- ▶ It allows the comparison of operating system designs;
- ▶ It allows control of experimental conditions (reduction of variance) that would only be possible through experimentation;
- ▶ It allows the compression of long operations, or even the expansion of time.

Disadvantages [Banks, 2005, Law et al., 2007]

- ▶ Stochastic simulation produces estimates, which depend on several rounds;
- ▶ If the characteristics of a problem can be estimated by exact parameters, optimization is preferable;
- ▶ Software can be expensive and model development is time-consuming;
- ▶ Realistic animations can be impressive, but there is a risk that the model may not be valid (Type Zero Error);

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- ▶ **Integration limitations:** Software programs are limited to vendors.
 - Python: **diverse integration environment**. Statistical analysis libraries, machine learning, visualization, powerful data manipulation packages (pandas) can be combined in an integrated way with SimPy. **Strong integration**. There is no parallel in the world. There are many technologies from different fields of knowledge.

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- ▶ Maintenance and technology transfer:
 - On SimPy, version the code; document and adopt good programming practices. Python is a **global language**. It gives you control over the modeling process.

Python is good for simulation [Ceng, 2025]



1. Flexibility and control of the code;
2. Modeling with SimPy is intuitive and the code is clean (clear);
3. Access to the complete Python ecosystem (full of ready-made libraries);
4. Completely free. No barriers to use;
5. Transparent and reproducible models; (show the model to customers, if necessary)
6. Fast interaction and innovation;
7. Knowledge of Python is valuable. Knowledge only grows.

SimPy: Python's most popular discrete event simulation library

[Zinoviev, 2024, Ceng, 2025]



1. Powerful and elegant models, with the largest number of users and community (solutions and AI);
2. Easy to use and not a set of varied techniques. It is well limited to a clear objective: **DES**;
3. Highly recognized in the industry as well;
4. **Intuitive**: SimPy (event-based) models without having to go into details and abstractions;
5. **Powerful**: Solves simple to complex multi-process problems;
6. *Start small and evolve with more details*;
7. **Disadvantage*: No built-in animation (yet).

DESK: A Discrete Event Simulation Kit [Almeida, 2025]



1. Modular Block Architecture: Reusable building blocks (**CREATE, PROCESS, DECIDE, DISPOSE**);
2. Event Tracing: Event logging with filtering and replay;
3. Visualization: Animation interface synchronized with event log printing;
4. **DistFit** Tool: Automated distribution fitting with statistical tests (desk-distfit) ;
5. Replication Framework: (**Simulation**);
6. Factorial Experiments: Factorial design with interaction analysis.

Creating conceptual models [Ceng, 2025]

1. Use models (maps) as communication tools with process managers. E.g.: A figure, processes (1,2,3), resources, and a flowchart for each process (1,2,3).
 - Record assumptions, queues (FIFO), number of resources, capacity, times. (best case, worst case, and most likely)

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 - Alignment and initial negotiation with the information manager. (Success is in delivering what was promised)
 - Model qualification: What is included and **what is clearly excluded?**

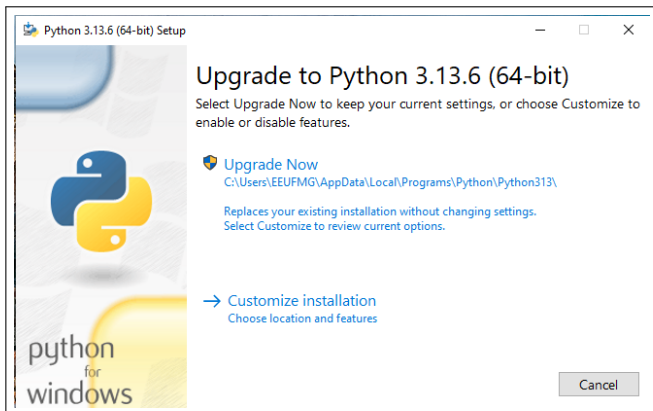
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3. Project vs. Product:
 - Project: study with defined goals and start and end dates (grows in complexity, automatic testing).
 - Product: Open goals that change, continuous development, and improvement.
 - MVP: Clear goals and limited time. = Project.

Softwares setup

Python: (Download latest version at: <https://www.python.org/downloads/>)

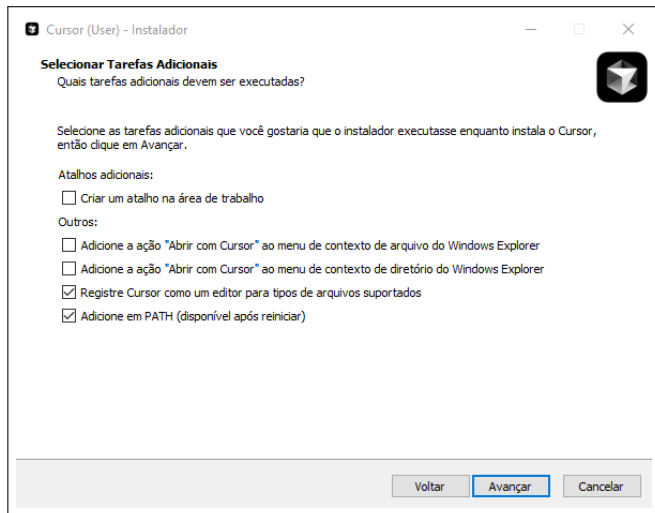
Important Option: Add to PATH variable.



Softwares setup

IDE: Cursor (or any other): (Download latest version at: <https://cursor.com/>)

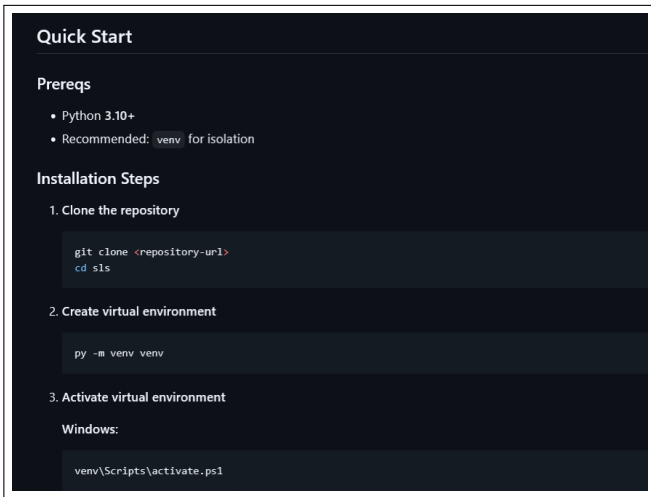
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Softwares setup

Codes: Github: <https://github.com/joaoflavioufmg/sls>

Follow README file.



The image is a screenshot of a README file with a dark background and light text. It is titled 'Quick Start' and contains sections for 'Prereqs' and 'Installation Steps'. The 'Prereqs' section lists Python 3.10+ and a recommendation to use 'venv' for isolation. The 'Installation Steps' section contains three numbered steps: 1. Clone the repository, 2. Create virtual environment, and 3. Activate virtual environment. Each step is followed by a code block containing the necessary commands. Step 1 shows 'git clone <repository-url>' and 'cd sls'. Step 2 shows 'python -m venv venv'. Step 3 shows 'venv\Scripts\activate.ps1' for Windows.

Quick Start

Prereqs

- Python 3.10+
- Recommended: `venv` for isolation

Installation Steps

1. Clone the repository

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git clone <repository-url>
cd sls
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2. Create virtual environment

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python -m venv venv
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- ▶ Download (or clone) *the course Github repository*;
- ▶ Create a virtual environment: `py -m venv venv`
- ▶ Activate the virtual environment: `. /venv/Scripts/activate.ps1` (**venv**)
- ▶ Install requirements: `pip install -r requirements.txt`
- ▶ Install ipython: `pip install ipython`
- ▶ Open 00-Supplementary-Material-Python-Course
- ▶ Run Ipython: `> ipython`
- ▶ Open a file to study: e.g. 01-Sintaxe.py
- ▶ Select each line (or more) and SHIF + ENTER

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